DeltaGrad: Towards Robust Deep Learning via Adaptive Gradient Reliability*

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1 Introduction

Modern adaptive optimizers like Adam [1] often "over-smooth" gradients, leading to noise memorization and instability. While previous methods such as SAM [2] refined this approach–improving generalization by smoothing the loss landscape–they incur high computational costs.
This paper presents **DeltaGrad** as a solution; it introduces a real-time *Reliability Metric* (R_t) to modulate updates via instantaneous gradient coherence. The present research aims to evaluate the feasibility of DeltaGrad as a robust alternative to traditional adaptive optimizers, while systematically characterizing its core operational behaviors.

2 Methodology

DeltaGrad introduces a self-regulating update mechanism based on gradient coherence. First, a lightly smoothed gradient $g_{smooth}^{(t)}$ is computed:

$$g_{smooth}^{(t)} = \sigma g_{smooth}^{(t-1)} + (1-\sigma)\nabla L^{(t)} \tag{1}$$

using a low-range factor $\sigma \in [10^{-4}, 0.1]$ to ensure stability gains originate from reliability logic rather than damping. The core innovation is the **Reliability Metric** (R_t):

$$R_t = \text{clamp}\left(\frac{K - \sum_{j=0}^{K-1} \alpha^j \Phi_j}{K}, 0.1, 1.0\right) \tag{2}$$

which quantifies the reliability of the signal against a window of $K-1$ previous gradients. Clamping prevents gradient paralysis. Computing a normalized distance Φ_j:

$$\Phi_j = \frac{|g_{smooth}^{(t)} - g^{(t-j)}|}{|g_{smooth}^{(t)}| + |g^{(t-j)}| + \epsilon} \tag{3}$$

Weighted by exponential decay α, R_t acts as a dynamic safety brake: it approaches 1.0 during coherent trajectories and scales down to 0.1 when noise dominates.

The update direction is governed by the **Windowed Inertia** ($\hat{m}_t$):

$$\hat{m}_t = \frac{\sum_{j=0}^{K-1} \beta^{j+1} \cdot g^{(t-j)}}{\sum_{j=0}^{K-1} \beta^{j+1}} \tag{4}$$

Unlike infinite-horizon EMAs, this finite memory remains responsive to local geometry.

*Official Repository: `https://github.com/xandasoneill/deltagrad_optimizer.git`

The final update integrates these components:

$$\theta_{t+1} = \theta_t - (\mu \cdot \gamma) \cdot (\hat{m}_t \odot R_t) \tag{5}$$

where μ is the learning rate and γ is a sensitivity coefficient. The product $\hat{m}_t \odot R_t$ allows for fine-grained dampening, suppressing oscillations that lead to divergence.

3 Experiments

Three experiments were conducted to evaluate the robustness of DeltaGrad: a Learning Rate Stress Test, a Batch Size Test, and a Data Noise Test. To provide a consistent baseline, a compact ConvNet architecture was trained using both Adam and DeltaGrad, with both optimizers tuned via Optuna [3] for an equal number of trials on the CIFAR-100 dataset[4]. This extended abstract focuses on the results from the Data Noise and Learning Rate Stress Tests; further results and the complete implementation are available in the official repository.

3.1 General Results

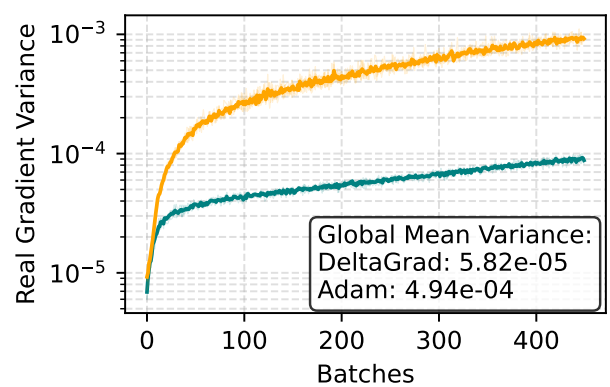

Figure 1: Gradient Variance Comparison(DeltaGrad-blue, Adam-orange) for batch size of 512 over 5 runs.

High gradient variance indicates that a neuron has not yet converged to a stable feature representation, often due to conflicting signals from labeled noise [5]. By prioritizing parameters that exhibit directional consistency, DeltaGrad effectively favors 'feature learning' over 'noise memorization'.
In all tests, DeltaGrad stabilizes gradient variance an order of magnitude below Adam [1], demonstrating that the reliability metric R_t (Eq. 2) effectively guides the optimizer toward more stable regions, ensuring the model learns stable data patterns instead of transient label errors.

3.2 Learning Rate Stress Test

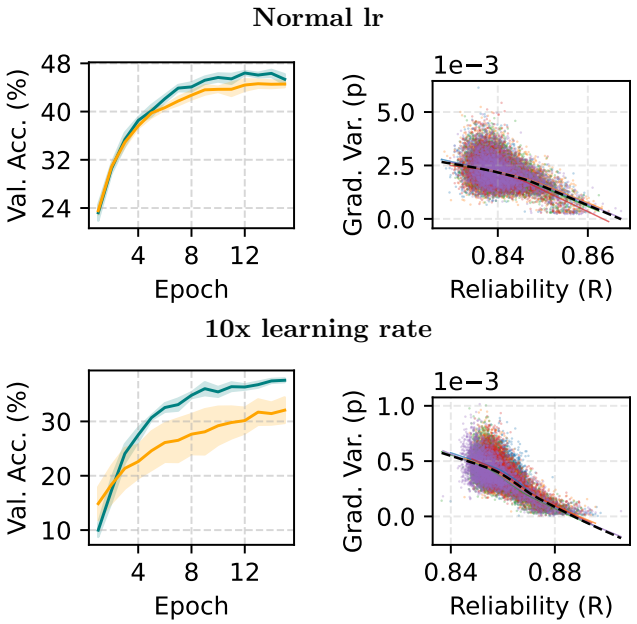

Figure 2: Validation accuracy (left) and R_t-gradient variance correlation (right) for Adam (orange) and DeltaGrad (blue). Solid lines and shading denote mean ± standard deviation over 5 runs across different learning rate regimes, with individual runs color-coded in the scatter plots (right).

Table 1: Validation accuracy standard deviation and mean R_t-gradient variance correlation (over 5 runs) for Adam and DeltaGrad across three learning rate regimes.

LR	Acc. Std. Dev. (%)		Avg. Corr. (Var. - R_t)
	DeltaGrad	Adam	
Normal	0.71	0.64	-0.473
3x	0.93	1.76	-0.599
10x	0.93	3.18	-0.709

As illustrated in Figure 2 and Table 1, DeltaGrad exhibits significantly higher consistency in validation accuracy across varying learning rate regimes compared to Adam [1]. Using a fixed batch size of 16 to intentionally amplify stochastic noise, the results show that the correlation between gradient variance and the reliability factor R_t strengthens with elevated learning rates.
Interestingly, despite the increased noise typical of higher learning rates, DeltaGrad maintains lower gradient variance in these regimes. This is accompanied by a normal accuracy learning curve that eliminates the gradient paralysis scenario [5]. These findings suggest that R_t functions as an effective damping mechanism, suppressing the high-frequency oscillations common in noisy landscapes [2] and ensuring stability even under aggressive optimization settings.

3.3 Data Noise Test

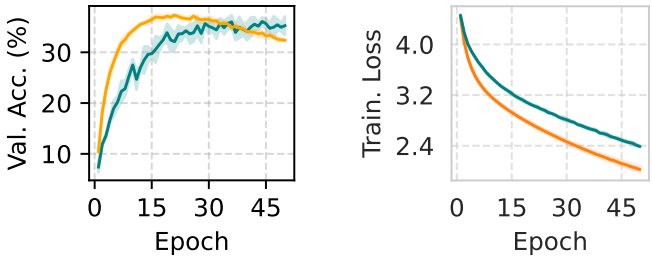

Figure 3: Validation accuracy (left) and training loss (right) under 20% label noise (batch size=512). Mean ± standard deviation (5 runs), noting that the standard deviation in accuracy is 0.53% for Adam(orange) and 1.42% for DeltaGrad(blue).

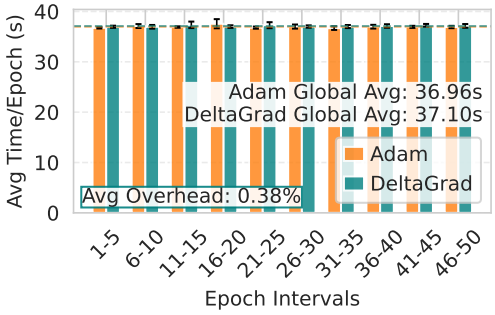

Figure 4: Average training time per epoch (batch size of 512; 20% data noise - randomly flipped labels).

Under 20% synthetic label noise(randomly flipped labels), DeltaGrad demonstrates superior robustness compared to Adam [1]. While Adam suffers from accuracy degradation after epoch 20 due to noise memorization—evidenced by its lower training loss—DeltaGrad maintains a stable upward trajectory, reaching a higher final accuracy. The higher training loss in DeltaGrad confirms the efficacy of the R_t metric in filtering inconsistent signals, prioritizing generalization over the minimization of noisy objectives [5]. This resilience is achieved with a negligible computational overhead of only 0.38%, suggesting that the mechanism is highly scalable.
The increased accuracy deviation observed with larger batch sizes (here, 512) stems from reduced R-update granularity and the heightened impact of Windowed Inertia ($\hat{m}_t$) over fewer update steps.

4 Conclusions

These outcomes position DeltaGrad as a robust framework for noise-resilient optimization in high-stakes production environments.
Despite these promising results, the current evaluation of DeltaGrad is subject to computational constraints that limit the initial testing scope. To address these limitations, future research will evaluate the optimizer across broader datasets and diverse neural architectures.

References

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